The later canonical-form answer behind arXiv:1507.00613

Literature-status note for `fa_banach_001`

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Status. `literature_already_answered`: canonical representation of isometric monoid isomorphisms for convex, bounded-below, 1-Lipschitz functions on Banach spaces.

The exact source question

For a Banach space X, let

$$\mathcal{CL}^1(X) = \{f : X \to \mathbb{R} : f \text{ is convex and bounded below, and } f \text{ is 1-Lipschitz}\},$$

equipped with infimal convolution and the bounded transform of the uniform metric. Bachir's 2014 paper asks whether every isometric monoid isomorphism between such function monoids has the canonical composition form induced by an isometric isomorphism of the underlying Banach spaces [1]. The 2015 target restates that uncertainty immediately before its Katetov-monoid Banach–Stone theorem: it says that the canonical form is not known for "the set of all convex 1-Lipschitz bounded from below functions defined on Banach space" and points explicitly to Problem 2 of the 2014 paper [2].

The later affirmative answer

In 2016 Bachir proved a complete canonical-form theorem for the larger nonnegative 1-Lipschitz monoids on invariant metric groups. In the notation of that paper, if

$$\Phi : (Lip^1_+(X), \oplus, \rho) \longrightarrow (Lip^1_+(Y), \oplus, \rho)$$

is an isometric monoid isomorphism, then there is an isometric group isomorphism $T : \overline{X} \to \overline{Y}$ such that

$$\Phi(f) = (\overline{f} \circ T^{-1})|_Y \qquad (f \in Lip^1_+(X)).$$

This is Theorem 1 of arXiv:1610.02825 [3]. For complete Banach spaces, the completions in that statement are just X and Y.

Most decisively, Remark 1(2) of the same paper states that following this proof and replacing "1-Lipschitz function" by "1-Lipschitz and convex function" gives a positive answer to Problem 2 of the 2014 paper. This is an explicit question-to-answer cross-reference, not an inference from related results.

The proof mechanism also explains why convexity causes no new ambiguity. The monoid units identify the Kuratowski functions and hence the underlying isometry T; isometry of the monoids recovers order and constant translations; every 1-Lipschitz function has the envelope

$$f = \inf_{t \in X} \{f(t) + \delta_t\};$$

and the isomorphism preserves this infimum. Applying the same argument inside the convex submonoid yields $\Phi(f) = f \circ T^{-1}$. Since an isometric group isomorphism between real Banach spaces fixes 0, the Mazur–Ulam theorem makes T linear, so this is the canonical Banach-space form requested in Problem 2.

Conclusion and scope

The sole live "we do not know" statement in arXiv:1507.00613 is therefore not a fresh open problem. It was answered affirmatively in arXiv:1610.02825, with an explicit citation to the original problem. The second scanner hit in the target is inside a commented-out TeX line about classifying continuous monoid morphisms to $\mathbb{R}$; it is not a published open statement and is excluded from the mathematical scope.

References

[1] M. Bachir, *The inf-convolution as a law of monoid. An analogue to the Banach–Stone theorem*, J. Math. Anal. Appl. 420 (2014), 145–166, https://doi.org/10.1016/j.jmaa.2014.05.074.

[2] M. Bachir, *Any law of group metric invariant is an inf-convolution*, arXiv:1507.00613, https://arxiv.org/abs/1507.00613.

[3] M. Bachir, *On representations of isometric isomorphisms between some monoid of functions*, arXiv:1610.02825, https://arxiv.org/abs/1610.02825.